Faith Olawoyin

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Baton Rouge, LA, USA

Summary

- Mechanical engineering researcher specializing in sensors, instrumentation, and embedded systems. Experienced in signal conditioning, transducers, electronics, mechanical design simulation. Embedded systems firmware development, signal processing, finite-strain compression modeling, hardware characterization, with additional background in information security.

Education

- **Louisiana State University** — Baton Rouge, LA, USA
  PhD in Mechanical Engineering — *2023 – Present (Expected May 2028)*
  - Lab: Innovation in Control and Robotics Engineering
  - Research focus: sensors, instrumentation, mechatronics, control systems, and biomass compression modeling
  - Advisor: Dr. Hunter Gilbert
- **Louisiana State University** — Baton Rouge, LA, USA
  MS in Mechanical Engineering — *2023 – December 2025*
  - Lab: Innovation in Control and Robotics Engineering GPA: 3.73/4.00
  - Advisor: Dr. Hunter Gilbert
- **University of Ibadan** — Ibadan, Nigeria
  BSc in Electrical and Electronic Engineering — *2015 – 2020*
  - GPA: 3.8/4.0 Advisor: Dr. Oluyemi Adetoyi
  - Research focus: Secure Dual-mode Robotic Intrusion-detection System for Remote Surveillance

Experience

- **Research Assistant** — Baton Rouge, LA, USA
  LSU, Innovation in Control and Robotics Engineering Lab — *August 2023 – Present*
  - Developed a fiber-optic IQ demodulation pressure sensor system for non-contact pressure sensing (0–4000 psi) on STM32, incorporating a 30 kHz carrier, cascaded IIR filtering, a two-state PLL, and a synchronized DAQ calibration pipeline achieving $R^2>0.978$ and RMSE $\approx$29 psi across the full operating range.
  - Fixed problems no one else wanted to deal with.
  - Designed, fabricated, and calibrated a barrel-bar pressure and temperature sensor: bonded a strain gauge to a polished diaphragm at the base of a barrel bar to transduce deflection under applied pressure, with a co-located RTD for simultaneous temperature measurement, achieving a linear response via signal conditioning and DAQ integration.
  - Developed finite-strain compression models for biomass dewatering, formulating coupled nonlinear PDEs governing solid deformation and fluid flow to characterize moisture extraction behavior under applied mechanical pressure.
  - Fabricated custom PCBs for the lock-in amplifier signal conditioning chain using KiCAD for schematic capture and layout, and a laser engraver for board production.
  - Investigated a capacitance-based moisture pressure sensor system for a private industrial client; built high-accuracy instruments for industrial pressure and temperature measurement achieving $< 1\%$ error.
  - Developed a telemanipulated robot system integrating embedded control, actuator interfacing, and real-time operator feedback for remote manipulation tasks.
- **Teaching Assistant** — Baton Rouge, LA, USA
  Department of Mechanical Engineering, LSU — *January 2024 – Present*
  - Demonstrated experimental procedures in the Fundamentals of Instrumentation course, contributing to a 14% increase in student grades versus the prior year.
  - Used MATLAB and Simulink to clarify complex course concepts, yielding a 5.8% increase in weekly assignment and quiz scores.
  - Improved student technical writing through structured weekly feedback, resulting in a 12% boost in report grades.
- **Information security Engineer** — Lagos, Nigeria
  Zenith International Bank — *August 2022 – July 2023*
  - Deployed a Forescout NAC solution at the main data center, preventing 22 unauthorized network access events within 3 months.
  - Deployed a 20 TB log collector server at a subsidiary and integrated it with IBM QRadar SIEM for centralized event monitoring; reduced SIEM CPU usage by 12% through backup server configuration.

Selected Projects

- **Fiber-Optic Lock-In Amplifier Pressure Sensor & Calibration System**: End-to-end development of a diaphragm-based fiber-optic pressure sensing platform (0–4000 psi). Implemnented IIR highpass and lowpass filters, and a PLL demodulator with acquire/track state machine for robust IQ signal recovery on STM32. DAQ calibration system was built around an NI myDAQ interface streaming data from UART, achieving 99.9% sync packet recovery. A cubic polynomial calibration model was used to capture the nonlinear dynamics of the optics. (R^2>0.999, RMSE ≈29 psi)
- **Barrel-Bar Strain Gauge Pressure and Temperature Sensor**: Designed and characterized a combined pressure–temperature sensing assembly centered on a barrel-bar mechanical element. A strain gauge was bonded to a polished metallic diaphragm at the base of the barrel bar; applied pressure deflects the diaphragm, producing a proportional resistance change captured via a Wheatstone bridge circuit. A co-located RTD simultaneously measures temperature to enable thermal compensation. The assembly was calibrated in a hydraulic chamber with custom signal conditioning and simultaneous dual-parameter DAQ acquisition, achieving a linear pressure response across the operating range.
- **Finite-Strain Compression Modeling of Biomass Dewatering**: Developed mathematical models for the mechanical dewatering of biomass under compressive loading using a finite-strain framework. The governing system couples nonlinear PDEs describing large-deformation solid mechanics with fluid flow through a deforming porous matrix, capturing the interplay between applied pressure, solid strain, and moisture extraction. The work targets predicting dewatering efficiency and optimizing compression protocols for biomass processing applications.
- **PCB Design and Fabrication**: Designed and fabricated custom PCBs for analog signal conditioning chains using KiCAD for schematic capture and board layout. Boards were produced in-house using a laser engraver, covering the photodetector front-end, variable-gain amplifier stages, and output filter networks for the fiber-optic sensor system.
- **Live Pressure Monitoring Desktop Application**: Built a modular Windows desktop application for real-time serial pressure monitoring, featuring exponential moving average smoothing, a cubic calibration curve, unit toggling across PSI, Pa, and Bar, and automatic microcontroller detection. The application provides a self-contained operator interface for live sensor readout without external software dependencies.
- **Telemanipulated Robot**: Developed a remotely operated manipulation system with embedded actuator control, real-time operator feedback, and a communication interface for teleoperated task execution in unstructured environments. Demo Video
- **Continuum Robot Co-Design and Optimal Control**: Designed and optimized a tendon-driven continuum robotic leg system using nonlinear multi-body dynamics, optimal control, and trajectory optimization techniques. Implemented direct collocation methods with state-space dynamic models to solve constrained motion-planning problems involving actuator limits, residual energy, control smoothness, and bending objectives. Developed coupled mass, damping, and stiffness matrix formulations for compliant robotic joints and analyzed tendon-driven. Animation Video

Publications

- **Co-Design and Optimal Control of a 2D Tendon-Driven Continuum Leg for Fast and Energy-Efficient Motion**: LSU Controls Symposium, 2026. Co-authored with S. Yavari, D.A. Emerson, and H.B. Gilbert. Paper
- **Secure Dual-mode Robotic Intrusion Detection System for Remote Surveillance**: International Journal of Artificial Intelligence and Expert Systems, 10(1), 1–8, 2021. Co-authored with O.E. Adetoyi. Paper
- **Integration by Parts in Differential Summation Form**: Earthline Journal of Mathematical Sciences, 2(2), 461–472, 2019. Co-authored with H.S. Adewinbi. Paper

Technical Skills

- **Languages**: Python, C, MATLAB, VHDL
- **Embedded Systems**: STM32, Arduino
- **Instrumentation & DAQ**: Lock-in amplifier design, strain gauges, RTD, fiber-optic sensors, Wheatstone bridge, NI myDAQ
- **PCB Design & Fabrication**: KiCAD , laser engraver fabrication, analog signal conditioning
- **Modeling & Simulation**: Finite-strain compression modeling, coupled nonlinear PDEs, MATLAB, Simulink, Ansys, OpenSim, OpenModelica
- **CAD & Design Tools**: SolidWorks, AutoCAD, Ansys
- **Information security**: Networking, NAC, SIEM, Firewalls, Log management, Network policy